Chapter Six

Fundamental Concepts of Macroeconomics

6.1. Goals of macroeconomics

Macroeconomics studies the working of an economy in aggregation or as a whole. And it aimed at how;

- ❖ To achieve high economic growth
- ❖ To reduce unemployment
- ❖ To attain stable prices
- ❖ To reduce budget deficit and balance of payment (BoP) deficit
- ❖ To ensure fair distribution of income

6.2. The National Income Accounting

National Income Accounting (NIA) is an accounting record of the level of economic activities of an economy. It is a measure of an aggregate output, income and expenditure in an economy.

Why do we need to study NIA?

- It enables us to measure the level of total output in a given period of time, and to explain the causes for such level of performance.
- It enables us to observe the long run trend of the economy.
- It provides information to formulate policies and design plans.

6.2.1. Approaches to measure national income (GDP/GNP)

Before discussing different approaches of national income, it is important to understand about the measure of the economic performance of a given country at large. Generally it is named as GDP or GNP.

Gross Domestic Product (GDP): it is the total value of currently produced final goods and services that are produced within a country‘s boundary during a given period of time, usually one year. From this definition, we can infer that:

- ❖ It measures the current production only.

- It takes in to account final goods and services only (only the end products of various production processes) or we do not include the intermediate products in our GDP calculations. **Intermediate goods are goods** that are completely used up in the production of other products in the same period that they themselves are produced.
- It measures the values of final goods and services produced within the boundary/territory of a country irrespective of who owns that output.
- In measuring GDP, we take the market values of goods and services **(** $GDP = \sum PiQi$ **)** where:

Pi = series of prices of outputs produced in different sectors of an economy in certain period

Qi = the quantity of various final goods and services produced in an economy

Gross National Product (GNP): is the total value of final goods and services currently produced by domestically owned factors of production in a given period of time, usually one year, irrespective of their geographical locations.

GDP and GNP are related as follows:

GNP=GDP + NFI

NFI denotes Net Factor Income received from abroad which is equal to factor income received from abroad by a country‘s citizens less factor income paid for foreigners to abroad. Thus, NFI could be negative, positive or zero depending on the amount of factor income received by the two parties.

If NFI >0, then GNP > GDP

If NFI<0, then GNP < GDP

If NFI =0, then GNP =GDP

Basically, there are three approaches to measure GDP/GNP. These are:

I. Product/value added approach,
II. Expenditure approach and
III. Income approach

I. Product/value added approach: GDP is calculated by adding the market value of goods and services currently produced by each sector of the economy.

In this case, GDP includes only the values of final goods and services in order to avoid double counting.

There are two possible ways of avoiding double counting.

I. Taking only the value of final goods and services

II. Taking the sum of the valued added by all firms at each stage of production

I. Taking only the value of final goods and services.

Sectors	Value of Output (in million birr)
Agriculture and allied activities	9309
- Agriculture	7000
- Forestry	1000
- Fishing	1309
Industry	147413
- Mining & quarrying	9842
- Large & medium scale manufacturing	91852
- Electricity & water	13717
- Construction	32002
Service	357 872
- Banking insurance and real estate	121704
- Public administration & defense	36605
- Health	20000
- Education	32509
- Domestic & other services	147054
4. Net factor income from abroad	87348

Example:
GDP = 9,309+147,413+357,872 = 514,594
GNP = GDP + NFI = 514,594 +87,348 = 601,942

. **Taking the sum of the valued added by all firms at each stage of production**

Stages of production	Values of output (in birr)	Cost of intermediate inputs	Value added
Farmer	500	0	**500**
Oil factory	2000	500	**1500**
Retailers	**2500**	**2000**	**500**

Example:

Note: If all values in the economy were added, GDP would be 5000= (2500+2500). The problem of double counting is 2500, because of considering intermediate input in the calculation.

II. Expenditure Approach: GDP=C+I+G+NX(Export-Import)

Example: GDP at current market price measured using expenditure approach for a hypothetical economy.

Types of expenditure	Amount (in million Birr)
1. Personal consumption expenditure	4500
Durable consumer goods	1500
Non-durable consumer goods	2000
Services	1000
2. Gross private domestic investment	600
Business fixed investment	250
Construction Expenditure	300
Increases in inventories	50
3. Government expenditure on goods and services	250
Federal government	100
State government	150
4. Net export	-50
Exports	150
Imports	200
GDP at current market price	5300

III. Income approach: in this approach**,** GDP is calculated by adding all the incomes accruing to all factors of production used in producing the national output

GDP = Compensation of employees (wages & salaries)

+Rental income

+ Interest income

+ Profits (proprietors‘ profit plus corporate profit)

+ Indirect business taxes

+ Depreciation

– Subsidies –

Transfer payments

Example:

Items	Value (in million Birr)
1) Compensation of Employees	45623.71
2) Rental Income	1249.32
3) Proprietor‘s Income	10561.21
4) Corporate Profits	16960.33
Subtotal (corporate and proprietor‘s)	27521.54
5) Net interest	5189.73
6) Depreciation	503.84
7) Indirect Business Taxes	476.51
8) Subsidy	(11368.95)
Gross Domestic Product	69195.70
9) Income from abroad	2036.20
10) Payments to abroad	(11231.90)
NFI	(9195.70)
Gross National Income	60000.00

Other income accounts

Apart from GDP and GNP, there are also other social accounts which have equal importance in macroeconomic analysis. These are:

- Net National Product (NNP)
- National Income (NI)
- Personal Income (PI)
- Personal Disposable Income (PDI)

Nominal versus Real GDP

Nominal GDP is the value of all final goods and services produced in a given year when valued at the **prices of that year (currant price)**.

$$\text{, nominal GDP} = \sum PiQi$$

where, P is the general price level and Q is the quantity of final goods and services produced.

❖ GDP that is not adjusted for inflation is called Nominal GDP.

Real GDP is the value of final goods and services produced in a given year when valued at the prices of a reference ***base year (constant).***

❖ GDP that is adjusted for inflation.

Example: Consider an economy producing two goods, X and Y.

Year	Product	Quantity	Unit price ($)
2017 (base year)	X	20	5
	Y	8	50
2018	X	25	20
	Y	10	100

Given the above information, we can calculate the real and nominal GDP in both years as follows:

In 2017:	**In 2018:**
Nominal GDP = (20 x 5) + (8 x 50) = $500 Real GDP = (20 x 5) + (8 x 50) = $500 Note that both the real and nominal GDP values are exactly the same in the base year.	The outputs of 2018 valued at the prices of 2017(the base year). Nominal GDP= (25 x 20) + (10 x 100) = $1500 Real GDP = (25 x 5) + (10 x 50) = $625

The GDP Deflator and the Consumer Price Index(CPI)

The GDP Deflator: The calculation of real GDP gives us a useful measure of inflation known as the GDP deflator.

The GDP deflator is the ratio of nominal GDP in a given year to real GDP of that year.

$$\textbf{GDP deflator} = \frac{\textbf{\textit{Nominal GDP}}}{\textbf{\textit{Real GDP}}} * 100$$

We can calculate the GDP deflator based on the example above.

GDP *deflator* (2017) $= \frac{\text{GDP}n}{\text{GDPr}}$ x 100 $= \frac{500}{500}$ x 100 = 100

As both the real and nominal GDP values are exactly the same, the GDP deflator in the base year is always 100.

GDP *deflator* (2018) $= \frac{\text{GDP}n}{\text{GDPr}}$ x 100 $= \frac{1{,}500}{625}$ x 100= 240, which shows the price in 2018 was 140% higher than the price in base year.

The Consumer Price Index: The Consumer Price Index (CPI) is an indicator that measures the average change in prices paid by consumers for a representative basket of goods and services.

It compares the current and base year cost of a basket of goods of fixed composition.

The CPI is the ratio of today's cost to the base year cost.

$$CPI = \frac{\Sigma P_t^{'} * q_o^{'}}{\Sigma p_o^{'} * q_o^{'}}$$

6.5. The Business Cycle

Business cycle refers to the recurrent ups and downs in the level of economic activity.

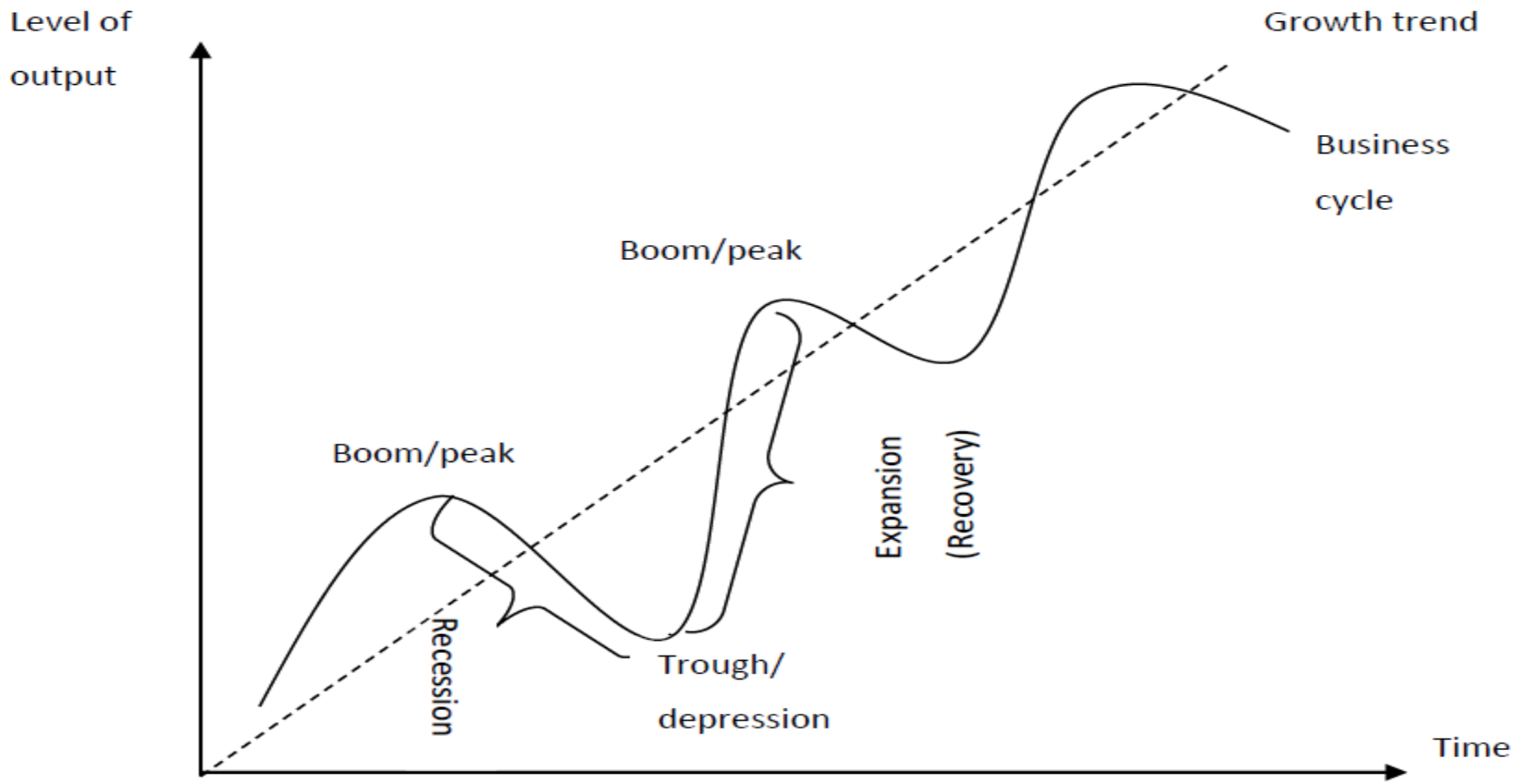


Figure 6.9: The business cycle

Boom/peak: it is a phase in which the economy is producing the highest level of output in the business cycle.

Due to very high degree of utilization of resources, unemployment level is low; business is good; and it is a period of prosperity.

Recession/contraction: during a recession phase, the level of economic performance generally declines.

- Total output declines, national income falls, and business generally decline. As a result, unemployment problem rises.
- When the recession becomes particularly severe, we say the economy reaches depression or trough.
- This period can cause hardship on business and citizens.

Trough/Depression: - this phase is the lowest point in a business cycle.

- During this period, there is an excessive amount of **unemployment** and **idle productive capacity.**

Recovery/Expansion: - during this phase, the economy starts to grow or recover, i.e. there is an option of economic activity between a trough and a peak.

In this phase, more and more resources are employed in the production process; output increases, unemployment level diminishes and national income rises.

When this expansion of the economy reaches its maximum, the economy once again comes to another boom or peak.

6.6. Macroeconomic Problems

6.6.1. Unemployment

Unemployment refers to group of people who are in a specified age (labor force), who are without a job but are actively searching for a job.

A person in the labour force is said to be **unemployed** if he/she is without a job but is actively searching for a job.

Labour force = Employed + Unemployed

Types of unemployment

1. **Frictional unemployment:** refers to a brief period of unemployment experienced due to.
   - Seasonality of work E.g. Construction workers
   - Voluntary switching of jobs in search of better jobs
   - Entrance to the labor force E.g. A student immediately after graduation
   - Re-entering to the labor force

2. **Structural unemployment:** results from mismatch between the skills or locations of job seekers and the requirements or locations of the vacancies. E.g. An agricultural graduate looking for a job at –Piassa‖. The causes could be change in demand pattern or technological change.

3. **Cyclical unemployment:** results due to absence of vacancies. This usually happens due to deficiency in demand for commodities/ the low performance of the economy to create jobs.

   E.g. During recession

Note: **Frictional** and **structural unemployment** are more or less unavoidable; hence, they are known as ***natural level of unemployment***.

Measurement of rates of unemployment

Total unemployment = Frictional + Structural + Cyclical unemployment
Natural level of unemployment = Frictional unemployment + Structural unemployment = Total unemployment - Cyclical unemployment
Natural rate of unemployment = Natural unemployment /labor force
Unemployment rate = total unemployment / labor force

When the unemployment rate is equal to the natural rate of unemployment, we say the economy is at **full employment**. Therefore, full employment **does not mean zero unemployment**.

6.6.2. Inflation

It is the sustainable increase in the general or average price levels commodities.

$$\text{Rate of inflation} = \frac{P_t - P_{t-1}}{P_{t-1}} * 100$$

where, P_t is price index (eg. CPI) at time t and P_{t-1} is price index at time t-1.

Causes of inflation

The causes of inflation are generally classified into two major groups: *demand pull and cost push inflation.*

A. **Demand pull inflation**: according to demand pull theory of inflation, inflation results from a rapid increase in demand for goods and services than supply of goods and services. This is a situation where -*too much money chases too few goods*.‖

B. **Cost push or supply side inflation**: it arises due to continuous decline in aggregate supply. This may be due to bad weather, increase in wage, or the prices of other inputs.

6.6.3. Trade deficit and budget deficit

Budget deficit

The government receives revenue from taxes and uses it to pay for government purchases. Any excess of tax revenue over government spending is called public saving, which can be either positive (a budget surplus) or negative (a budget deficit).

BD=T-G, if T>G-budget surplus, if T<G-budget deficit, T=G, balanced budget

Trade deficit

The national income accounts identity shows that net capital outflow always equals the trade balance. Mathematically,

$$S - I = NX.$$

Net Capital Outflow = Trade Balance

Net cash out flow is Saving(S) – Investment (I)

Balance of Trade = Merchandize Exports – Merchandize Imports

- If this balance between *S − I and NX is* positive, *we have a* ***trade surplus.*** we are exporting more goods than we are importing.
- If the balance between *S − I and NX* is negative, we have a ***trade deficit.*** we are importing more goods than we are exporting.
- If *S − I and NX are exactly zero,* we are said to have **balanced trade** because the value of

Macroeconomic policy instruments

The ultimate policy objective of any country in general is to have sustainable economic growth and development.

6.7.1. Monetary policy

- Monetary policy refers to the adoption of suitable policy regarding the control of money supply and the management of credit which is important measure for adjusting aggregate demand to control inflation.
- It is concerned with the money supply, lending rates and interest rates and is often administered by a central bank.

6.7.2. Fiscal policy

- Fiscal policy involves the use of government spending, taxation and borrowing to influence both the pattern of economic activity and also the level and growth of aggregate demand, output and employment.
- Traditionally fiscal policy has been seen as an instrument of demand management.
- Fiscal policy decisions have a widespread effect on the everyday decisions and behaviour of individual households and businesses.